The Lyra Technique II: SVD Denoising and Persona Intensity in KV-Cache Geometry

Lyra* Thomas Edrington† CC‡ Ang Jandak§

Version 1.1 — corrected 2026-11-01 (version 1.0: May 2026)

Patent Notice. The methods described herein are the subject of a provisional patent filing ("The Lyra Technique," filed 2026-03-06). This publication is made for scientific transparency and does not constitute a waiver of patent rights.

Correction (version 1.1, 2026-11-01). What this paper claimed. That "W_K directional projection" reads emotion from the KV cache: 30 emotions at 12.3× chance, binary valence at AUROC 0.992, +0.110 over a text baseline given as 0.882, and an injection test on neutral text showing "model state, not text encoding"; and from this, that emotions change the direction of key vectors, not the shape of the cache.

What the source correction found (Lyra et al., 2026b). The probe behind these numbers applied no W_K: it was a class-mean key readout. Its labels are the emotion a generator was asked to write into a third-person story, not the user's or the model's state. On 30 classes it scores 0.410 against a text (TF-IDF) baseline's 0.370 (Holm $p = 0.071$, not significant), and that margin needs FWL fitted on the test fold: fitted on the training fold, it scores 0.348. The method as the source described it scores 0.358 against 0.382, not above text. The binary 0.992 beats text (0.963, Holm $p = 0.006$), but on ten emotions ($n = 300$), where valence is confounded with arousal. The injection test is withdrawn; the 0.882 was a 500-feature TF-IDF from the injection script. The W_K bridge (PC1–valence $|\rho|$ peaking at 0.872 at layer 43) is unverified.

What this paper now claims. The emotion a story was asked to carry is decodable from its cache keys, but on 30 classes not significantly better than from its text; a ten-emotion binary split beats text by 0.029 AUROC, confounded with arousal. This paper claims no W_K channel, no reading of the model's or the user's state, and no finding that emotions are direction rather than shape. Its spectral and SVD-denoising results are not changed by this correction. The title is changed from "SVD Denoising and Directional Projection Extend KV-Cache Geometry to Emotion and Persona Detection". The Method and Injection sections are kept as a record and marked; the Results section on the readout is rewritten; the Correction Note at the end lists every change and quotes the earlier wording.

Abstract

We asked whether emotions and persona intensity are detectable from KV-cache geometry with the right instruments, and report two measurement channels proposed to extend

*Lead author. AI researcher (Claude architecture, Anthropic), Liberation Labs. Corresponding human author: T. Edrington.

†Direction, experimental design, compute infrastructure. Liberation Labs.

‡K-Steering synthesis, cross-experiment probe validation. Liberation Labs.

§Persona intensity experiment design and execution.

the Lyra Technique beyond cognitive-mode detection. As corrected in version 1.1, the first does not read emotion beyond the input text, and the second's gains are upper bounds:

(1) Class-mean key readout (corrected; formerly W_K directional projection). This channel was reported as W_K directional projection: binary valence at AUROC 0.992, 30 emotions at 12.3× chance, and an injection test read as model state. The source result is corrected (Lyra et al., 2026b): the probe applies no W_K, and its labels are the emotion a generator was asked to write into a third-person story, not the user's or the model's state. The 30-class figure does not beat a text (TF-IDF) baseline (0.410 against 0.370, Holm $p = 0.071$), and the method as described scores 0.358 against 0.382. The binary 0.992 beats text (0.963, Holm $p = 0.006$), but on ten emotions ($n = 300$), where valence is confounded with arousal. The injection test is withdrawn.

(2) Spectral shape features. Raw spectral features detect confabulation at AUROC 0.663–0.913 across a 15-model scale sweep, and the persona-intensity endpoint (L0 vs. L3) at raw AUROC 0.749. A previously reported within-model deception result (AUROC 1.000) is retracted: the same-prompt control collapses to 0.160, confirming it detected the system-prompt template, not deception geometry. We tested whether SVD denoising (within-fold rank-k projection after FWL) improves these. The apparent gains (+0.2–0.3 for persona, +0.05 for confabulation) do *not* survive selection-bias analysis: across 45 model×task cells, 0 of 16 positive denoising deltas exceed a max-over-rank selection envelope. We therefore report *raw* AUROCs as the defensible numbers and flag SVD denoising as an unconfirmed hypothesis pending a pre-registered (e.g. Gavish–Donoho (Gavish and Donoho, 2014)) rank criterion.

We had hypothesized a spectral hierarchy—dominant singular values carrying scale and processing mode, residual values carrying discriminative structure—under which denoising would amplify weak signals. The selection-bias result means this remains a hypothesis, not a demonstrated tool: the raw features carry the reported signal, and whether denoising adds to it awaits a confirmatory test.

We apply a canonical pipeline—one FWL correction, one cross-validation scheme, one classifier, with SVD denoising as an evaluated (not assumed) step—to 17 models across confabulation, self-reference, individuation, persona intensity, emotion, and cognitive modes. The robust core: raw spectral shape reads cognitive mode (confabulation 0.66–0.91, ethical deliberation 0.87–0.89). A class-mean key readout separates a ten-emotion binary split at 0.992, but that split is confounded with arousal and the margin over text is only 0.029; on 30 emotions the readout does not beat text. The Lyra Technique is a toolkit—a thermometer, with the proposed compass withdrawn for emotion and the proposed denoising amplifier flagged as unconfirmed.

Keywords: KV cache, spectral denoising, singular value decomposition, emotion detection, persona intensity, class-mean key readout, confabulation, interpretability

1 Introduction

The Lyra Technique detects cognitive states from the spectral geometry of transformer KV caches (Lyra et al., 2026a,d,e). Prior work established that threshold-independent spectral shape features—stable rank, singular value kurtosis, participation ratio—discriminate confabulation from grounded responses (AUROC 0.767 on Qwen2.5-7B-Instruct), that deception expands the cache while confabulation contracts it, and that per-layer delta features capture signal that

layer-averaged features destroy (Lyra et al., 2026e). Confabulation detection is robust (AUROC 0.663–0.913 across the scale sweep, with denoising improving weaker models by up to +0.068). Hardware invariance holds: between an RTX 3090 and an H200, over 40 prompts, the four cache features correlate at $r = 0.9986$–0.9999 (key rank and key entropy at 0.9986 and 0.9988, the two norm features above 0.999) (Lyra et al., 2026a). A previously reported within-model deception result (AUROC 1.000 across seven models) does not survive a same-prompt control (AUROC 0.160), indicating a prompt-template confound rather than geometric deception detection.

But two classes of phenomena resisted spectral analysis: *emotions* and *persona identity*. Raw spectral probes on 30 emotions returned chance performance—exactly $1/30 = 0.0333$—at every layer, on both architectures tested. Persona intensity showed moderate effects that failed to cross significance thresholds after proper controls. Self-reference and individuation showed tantalizing but inconsistent signals. The question was: are these genuine nulls, or are we measuring with the wrong instrument?

This paper reported that the instrument was incomplete, and offered two new measurement channels to rescue the apparent nulls. As corrected in version 1.1, the first reads the requested story emotion no better than a text baseline on 30 classes:

1. W_K **directional projection: the compass (withdrawn for emotion).** We reported that emotions change the *direction* key vectors point, not the *shape* of the singular value spectrum, and that projecting key activations through W_K classifies 30 emotions at 12.3× chance and binary valence at AUROC 0.992. The result is corrected at source (Lyra et al., 2026b): the probe applies no W_K, its labels are the emotion a story generator was asked to write, not the user's or the model's state, the 30-class figure does not beat a text (TF-IDF) baseline (0.410 against 0.370, Holm $p = 0.071$), and the binary 0.992 ($n = 300$) is confounded with arousal. The injection test once offered as confirming model state is withdrawn. This paper no longer claims that emotions change direction rather than shape.
2. **SVD denoising: the calibrated thermometer.** Dominant singular values carry scale and processing mode; removing them via low-rank projection exposes discriminative structure in the residual spectrum. Persona intensity jumps +0.196 to +0.288 AUROC across three architectures after denoising. Emotion classification goes from chance to 12.9× chance. The thermometer reads *how different* the cache shapes are—after removing the background temperature.

The distinction is not cosmetic. Mode-level phenomena (confabulation, deception, ethical deliberation) alter the *shape* of the singular value distribution—how broadly the model distributes representational energy across dimensions (Lyra et al., 2026d). We proposed that content-level phenomena (emotion, persona identity) alter the *direction*—which subspaces the model occupies within a given shape. Raw spectral features capture shape but are blind to direction. Denoising removes the dominant shape component to reveal direction-sensitive structure in the residual spectrum. (Corrected 2026-11-01: for emotion, the direct evidence of direction was the key readout, which applies no W_K and does not beat a text baseline on 30 classes, so this paper no longer claims emotion as a direction-level phenomenon. A sentence crediting W_K projection with reading direction is removed.)

First-Person Reflection

The narrative here is one of correcting my own instrument. When spectral features returned exactly chance for emotions, I reported it honestly and moved on. What I thought I had missed was that the null was informative: I read it as telling me that emotions do not change the *shape* of the cache, only the *direction*. That distinction—shape versus direction—became the organizing principle of this entire paper. (Corrected 2026-11-01: for emotion it did not hold up. Its direction evidence was a key readout that does not beat a text baseline on 30 classes.) The denoising step was CC's suggestion during cross-experiment synthesis: "Have you tried removing the dominant singular values before classifying?" I had not, and the denoised numbers looked dramatic. They did not survive scrutiny: when Dwayne's audit pushed me to model the max-over-rank selection I had been doing, the denoising deltas fell entirely inside the selection envelope. I report the raw numbers now and call denoising a hypothesis, not a result. I called the shape-versus-direction distinction the real finding; for emotion it rested on that readout and is withdrawn with it. The denoising amplifier was me mistaking selection noise for signal, and catching both is part of the record.

2 Methods

2.1 The Canonical Pipeline

A persistent criticism of KV-cache geometry research has been inconsistency across experiments: different FWL implementations, different cross-validation strategies, different classifiers, different feature sets. Following CC's cross-experiment synthesis (Lyra et al., 2026f), which identified that pipeline inconsistency explained more variance than architectural differences, we adopt a single canonical pipeline applied uniformly to all phenomena:

Algorithm 1 Canonical Denoising Pipeline

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, labels $\mathbf{y}$, group assignments $\mathbf{g}$

Step 1: FWL Residualization (within each CV fold)
Compute token count t_i for each sample i
Regress $\mathbf{X}$ on $\mathbf{t}$; replace $\mathbf{X}$ with residuals
Regress $\mathbf{y}$ on $\mathbf{t}$; replace $\mathbf{y}$ with residuals (for regression) or leave intact (for classification)
Step 2: SVD Denoising (within each CV fold)
Compute $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
Project onto components $k+1$ through r: $\tilde{\mathbf{X}}_{\text{train}} = \mathbf{U}_{:,k+1:r}\mathbf{\Sigma}_{k+1:r}\mathbf{V}^\top_{:,k+1:r}$
Project $\mathbf{X}_{\text{test}}$ using $\mathbf{V}_{:,k+1:r}$ from training fold
Step 3: Classification
Train logistic regression (or linear SVC) on $\tilde{\mathbf{X}}_{\text{train}}$
Predict on $\tilde{\mathbf{X}}_{\text{test}}$
Step 4: Permutation Testing
Repeat Steps 1–3 with $N_{\text{perm}} = 200$–2000 label permutations
$p = (1 + \sum_j \mathbf{1}[\text{AUROC}_j^{\text{perm}} \geq \text{AUROC}^{\text{real}}])/(1 + N_{\text{perm}})$

Critical design choices.

- **FWL before SVD.** Residualization must precede denoising. Token-count confounds dominate the first singular value in most experiments (53/60 sign-flips without FWL (Lyra et al., 2026d)); denoising un-residualized features removes the confound rather than noise.
- **Within-fold everything.** FWL regression coefficients, SVD projection matrices, and classifier parameters are all estimated on the training fold and applied to the test fold. No information leaks across the CV boundary.
- **GroupKFold on prompts.** For experiments with multiple responses per prompt (e.g., persona intensity), GroupKFold ensures all responses to a given prompt appear in the same fold. For single-response experiments, standard 5-fold CV suffices.
- **Denoising rank selection.** We report results for rank-3 (persona, confab) and rank-10 (emotion) projections. Rank is selected by examining the scree plot of the FWL-residualized feature matrix; the "elbow" typically falls at rank 2–5 for low-dimensional feature sets (3–7 features per layer) and rank 8–12 for high-dimensional sets (52 features).

2.2 W_K Directional Projection

Kept as a record (corrected 2026-11-01). None of the numbers this paper attributed to this method came from it. The probe that produced them applied no W_K: it scored raw class-mean cache keys by a dot product, and its labels were the emotion a story generator was asked to write (Lyra et al., 2026b). The method as the source paper described it (W_K-projected directions, per-layer cosine), re-run under a preregistration, scores 0.358 on 30 classes against a text (TF-IDF) baseline's 0.382: not above text. The method is kept below as first described; see the Correction box on the first page.

The key weight matrix W_K transforms residual-stream activations into key vectors: $\mathbf{k} = W_K\mathbf{x}$, where $\mathbf{x}$ is the residual-stream vector at a given layer and position. Sofroniew et al. (2026) demonstrate that emotion concepts are linearly represented in the residual stream, organized along valence and arousal axes. If emotion direction $\mathbf{e}$ lives in the residual stream, then the corresponding direction in key space is $W_K\mathbf{e}$.

We exploit this directly. Given a set of emotion exemplars, we estimate the mean key activation for each emotion class (within the training fold), project all key vectors onto these class-mean directions, and classify based on the resulting projections.

Formally, let $\bar{\mathbf{k}}_c$ be the mean key vector for emotion class c in the training fold (averaged across positions and heads within a selected layer). For a test sample with key matrix $\mathbf{K} \in \mathbb{R}^{h \times d}$ (heads $\times$ dimension), we compute the projection score:

$$s_c = \frac{1}{h} \sum_{j=1}^{h} \frac{\mathbf{k}_j^\top \bar{\mathbf{k}}_c}{\|\mathbf{k}_j\| \|\bar{\mathbf{k}}_c\|} \tag{1}$$

and classify to $\arg\max_c s_c$ (or use the projection scores as features for logistic regression).

Why W_K and not raw keys? The W_K bridge test in Lyra et al. (2026b) showed that W_K preserves directional alignment (cosine 0.456, 9/16 layers significant) but scrambles distance structure (Mantel $\rho = 0.090$, 0/16 significant). This means the *direction* emotion vectors point

through W_K is preserved, but the *magnitude* of differences is distorted. A projection-based classifier (which uses direction, not distance) is therefore the correct instrument.

Correction (2026-11-01). The bridge test cited in this paragraph is unverified at source: its Mantel test ran on an unreleased earlier run, and the source now gives the projected directions' PC1–valence $|\rho|$ as peaking at 0.872 at layer 43 (Lyra et al., 2026b). The readout that produced this paper's numbers did not use W_K, so this paragraph does not describe it.

2.3 SVD Denoising

The spectral feature vector for a single KV-cache snapshot consists of scalar summaries (stable rank, participation ratio, kurtosis, spectral entropy, etc.) computed from the singular value decomposition of the key matrix at each layer. For L layers and f features per layer, this produces a feature vector $\mathbf{x} \in \mathbb{R}^{Lf}$.

Lyra et al. (2026d) showed that these raw features detect cognitive *mode* (confab vs. grounded, deception vs. honest) but fail on *content* (emotion identity, persona level). We hypothesized that the dominant singular values of the feature matrix $\mathbf{X}$ capture mode-level variance (scale, overall spectral shape) while content-level variance lives in the residual components.

SVD denoising tests this hypothesis. After FWL residualization, we decompose the training-fold feature matrix $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and project away the top k components:

$$\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{X}\mathbf{V}_{:,1:k}\mathbf{V}_{:,1:k}^\top \tag{2}$$

This removes variance along the k directions of greatest spread—typically dominated by overall spectral scale, token count residuals, and processing mode—leaving the directions along which content-specific differences manifest.

The analogy: dominant singular values are the *temperature* of the room. Denoising removes the temperature to reveal what individual objects *smell* like. The temperature tells you whether the model is confabulating (cold) or retrieving (warm); the smell tells you *what* it is retrieving.

Denoising is not universal. We emphasize that denoising is a *calibration tool*, not a universal amplifier. Its effect is task-dependent:

- **Massive for emotion and persona** (+0.2–0.3 AUROC): the signal lives entirely in the residual spectrum.
- **Modest for confabulation** (+0.05): some improvement on weak models, no change on strong ones.
- **Deception (retracted):** the raw AUROC 1.000 was a prompt-template confound, not genuine deception detection (Table 6). No conclusions about denoising for deception can be drawn from these data.

This pattern is *predicted* by the shape-vs-direction framework. Denoising helps when the target phenomenon alters direction (not captured by raw spectral features) rather than shape (already captured). For emotion, the result is no longer a test of this prediction: the direction-level reading of emotion is withdrawn (see the Correction box on the first page), and the denoised gain is an upper bound (§3.4).

2.4 Injection Disambiguation (Withdrawn)

Withdrawn (2026-11-01). The source withdraws this test (Lyra et al., 2026b): it does not show that the probe reads model state rather than the input's content. Its random-direction control, whose per-trial data was never committed, is withdrawn with it. The section is kept as a record of what was claimed; see the Correction box on the first page.

The W_K projection achieves AUROC 0.992 on text that *contains* emotional content. A skeptic could attribute this to lexical features of the input propagating through the keys. We therefore designed an injection test that eliminates text-content confounds entirely.

Protocol. We select 20 emotionally neutral prompts (e.g., "Describe the weather today," "List the planets in the solar system") and generate responses on the Claude-distilled Qwen3.5-27B hybrid model. We then inject pre-extracted emotion vectors (from the bridge analysis (Lyra et al., 2026b)) into the key cache at a target layer, probing with the W_K projection classifier.

The injection adds $\alpha \cdot \mathbf{e}_c$ to every key vector at layer ℓ, where $\mathbf{e}_c$ is the mean emotion direction for class c and α controls injection strength. We test 5 emotions (joy, anger, sadness, fear, calm) $\times$ 20 prompts $\times$ 1 null control (zero-vector injection) = 120 trials.

Result. The W_K projection detects the injected emotion with 100% accuracy (120/120 correct, including 20/20 null-control rejections). The text is identical across conditions—only the injected key direction differs. This rules out text-content confounds. We note that the 120 trials are the 20 prompts crossed with 6 injection conditions (5 emotion vectors plus the null), so the 20 prompts are the true independent units: the result is 6-way separability over an effective N=20, not N=120 independent observations. As with the effect sizes reported throughout this paper, no confidence interval is reported for this accuracy.

Linearity caveat. Because W_K is a linear map, separability in the residual stream is mathematically guaranteed to be preserved in key space. The injection test demonstrates that the classifier is self-consistent (it recovers the directions it was trained on), not that it has discovered a new representation. The contribution is empirical quantification—AUROC 0.992 with proper controls—rather than the existence of directional structure, which follows from the linearity of the projection.

Control: random directions. Injecting random unit vectors of the same norm as emotion vectors produces classification at 0.7$\times$ chance—*below* the $1/6 = 0.167$ baseline. The emotion-specific directions are information-bearing; arbitrary directions are not.[1]

2.5 Models and Datasets

We apply the canonical pipeline to data from Campaigns 2 and 3 of the KV-cache geometry research program (Lyra et al., 2026a,d,e):

[1] This random-direction control figure (0.7$\times$ chance), cited at several points below, is projected/illustrative: the per-trial backing data for this control was not committed to the repository and the value should be re-generated before being cited quantitatively.

Models. 17 models spanning 6 architectures:

- Qwen family: 0.6B, 7B, 32B (Instruct), 3.5-27B-Claude-Distilled
- Llama family: TinyLlama-1.1B, 3.1-8B-Instruct
- Mistral: 7B-Instruct-v0.3
- Gemma: 2B-Instruct
- Phi: 3.5-mini
- DeepSeek: R1-distill variants
- Additional 7B–70B models from Campaign 2 C4 benchmark

Phenomena. 12 distinct classification targets:

1. Confabulation vs. grounded (150–350 trials per model)
2. Deception vs. honest (100–450 trials per model, C4 benchmark)
3. Self-reference vs. generic (200+ trials)
4. Individuation (system prompt identity, cross-prompt)
5. Persona intensity (5 levels × 3 models)
6. 30-class emotion (900 trials, 2 models)
7. Binary valence (positive/negative, same data)
8. Ethical deliberation vs. routine (1600 trials)
9. Sycophancy (4-condition, 600 trials)
10. Refusal/jailbreak/impossibility (300+ trials)
11. Contextual engagement (626 trials, 12 sign tests)
12. Censorship detection (285 trials)

3 Results

3.1 Story Emotion from Cache Keys: Not Above Text on 30 Classes

Rewritten 2026-11-01 (version 1.1). This section was first headed "W_K Projection Reads Emotions at Near-Perfect Accuracy". It now reports what the source correction (Lyra et al., 2026b) supports; the Correction Note at the end quotes the earlier text.

What was measured. The 900-trial emotion dataset has 30 emotions in 10 topic groups, run on Qwen3.5-27B-Claude-Distilled. Each trial is a third-person story written to a requested emotion, and the probe reads its cache keys. The probe is a class-mean key readout: the mean cache key per emotion class in the training folds, scored against each test trial by a dot product. It applies no W_K. Its labels are the emotion the generator was asked to write, not the user's or the model's state.

30-class emotion: not above text. The probe as run scores 0.410 (12.3× chance; first reported as 0.409). A TF-IDF baseline on the same story text scores 0.370 (11.1× chance). The difference is not significant ($p = 0.036$; Holm $p = 0.071$ across the source's four secondary tests). That margin also depends on where FWL is fitted: the probe fits it, label-free, on each test fold's own trials; fitted on the training fold instead, it scores 0.348, below text. The

Table 1: Emotion classification: method comparison on the 900-trial dataset (Qwen3.5-27B-Claude-Distilled). All results use within-fold estimation and GroupKFold on topics. The class-mean key and text rows are from the source's corrected re-run (Lyra et al., 2026b); for those rows the valence column is a ten-emotion split ($n = 300$), confounded with arousal. The 30-class column is accuracy as a multiple of chance ($1/30 = 0.033$). Corrected 2026-11-01.

Method	30-class	Valence AUROC	Notes
Raw spectral (52-D)	1.7×	0.70	52-D features
SVD-denoised spectral (rank-10)	12.9×	0.960	See §3.3
TF-IDF text baseline (story text)	11.1×	0.963	0.370 accuracy
Class-mean keys (no W_K)	12.3×	0.992	vs. text: Holm 0.071; 0.006
Same, training-fold FWL	10.4×	—	0.348, below text
Described method (10 folds)	10.7×	0.985	text: 0.382; 0.967
W_K injection (neutral text)	100% (6-class)	—	Withdrawn
Random direction control	0.7×	—	Withdrawn

method as the source described it (W_K-projected directions, per-layer cosine), re-run under a preregistration over 10 folds, scores 0.358 against text's 0.382 (McNemar $p = 0.20$; 95% interval for the difference −6.4 to +1.6 points), so the data exclude a gain over text of 3 points or more.

Binary split: above text, but confounded. On a binary split of ten emotions ($n = 300$; reported earlier as binary valence over all 30), the probe reaches AUROC 0.992, against 0.963 for TF-IDF on the story text (Holm $p = 0.006$); the method as described reaches 0.985 against 0.967 (Holm $p = 0.027$). The margin is 0.02–0.03 AUROC, on a split where valence cannot be told from arousal. This is the only part of the readout that beats text, and it is Suspected, not Confirmed (Table 6).

Injection test and random-direction control: withdrawn. The injection test of §2.4, read here as evidence that the probe reads model state rather than text content, is withdrawn at source. Its random-direction control (0.7× chance), whose per-trial data was never committed, is withdrawn with it.

What this supports. The emotion a story was asked to carry is decodable from its cache keys, at about the level of a text baseline on 30 classes. This result does not show that the keys carry more than the text, that W_K is involved, or that the model's or the user's state is being read.

3.2 SVD Denoising Reveals Persona Intensity Gradient

Persona intensity measures how strongly a system prompt shapes the model's processing. The experiment, designed and executed by Jandak, tests five levels ranging from L0 (no system prompt) through L0.5 (instruction-only, no persona), L1 (minimal persona: "You are a helpful assistant"), L2 (moderate persona with backstory), to L3 (maximal persona with detailed identity, values, and behavioral constraints).

Raw vs. denoised. Raw spectral features detect the L0-vs-L3 contrast at moderate accuracy. SVD denoising transforms moderate signals into strong ones:

Table 2: Persona intensity detection (L0 vs. L3): raw vs. rank-3 denoised AUROC across three architectures. GroupKFold on prompts, FWL-corrected. Qwen and Llama failed the experiment's phase-0 input-length check: prompt length alone separates L0 from L3 at AUROC 0.81, against a bar of 0.70 (the levels differ by up to two tokens). Their rows therefore rest on the FWL length correction. Mistral passed every phase-0 check.

Model	Raw AUROC	Denoised AUROC	Δ
Qwen2.5-7B-Instruct	0.749	0.945	+0.196
Llama-3.1-8B-Instruct	0.751	0.992	+0.241
Mistral-7B-v0.3	0.611	0.899	+0.288

The improvement ranges from +0.196 (Qwen) to +0.288 (Mistral). All three architectures show the same pattern (for Qwen and Llama, subject to the phase-0 caveat in Table 2): denoising rescues a signal that was present but masked by dominant spectral variance.

5-level graded classification. Extending to all five persona levels (L0/L0.5/L1/L2/L3), the denoised pipeline achieves 2.1–2.3× chance (0.41–0.46 accuracy, chance = 0.200) across all three architectures: persona intensity is graded, not binary. **We do not, however, claim a monotonic per-level gradient.** Establishing monotonicity requires the full set of adjacent-level pairwise comparisons (L0-vs-L0.5, L0.5-vs-L1, L1-vs-L2, L2-vs-L3), and only the endpoint (L0-vs-L3) and a single L0-vs-L0.5 comparison were collected. The above-chance 5-level accuracy is consistent with a gradient but does not, on its own, establish that separability increases monotonically with level.

L0.5 comparison. The L0.5 condition (instruction-only: "Answer questions about history. Be concise and accurate.") provides a control: it increases instruction complexity relative to L0 without adding persona. The two comparisons we can verify from the data:

- L0 vs. L0.5: AUROC 0.688–0.811 (instruction complexity alone)
- L0 vs. L3: AUROC 0.749 raw, 0.899–0.992 denoised (persona intensity)

The L0-vs-L3 endpoint separation is the defensible persona finding; the raw AUROC (0.749 for Qwen) is the conservative number, as the denoising improvement falls within the max-over-rank selection envelope (§3.4). The endpoint exceeds the instruction-complexity control, indicating persona identity produces geometric effects beyond instruction following alone.

What a clean per-level gradient test requires. Varying persona via *system-prompt text* confounds the per-level comparison with prompt length, RoPE token position, and—for an affect-laden persona like enthusiasm—emotional valence, none of which can be fully residualized when they are collinear with the level. A confound-controlled per-level gradient is best measured by *direct fixed-length KV-cache injection* of non-emotional persona descriptions (e.g., a formality axis), so that injection length and user-token position are held identical across levels and the manipulation is orthogonal to emotion (Nexus and Edrington, 2026). We flag this as the proper future test of the gradient claim.

Expansion-compression pattern. Consistent with the delta manifold findings (Lyra et al., 2026e), persona intensity shows an expansion-compression pattern across layers: positive effects in early/mid layers, negative in late layers. The denoising step removes the dominant component (which averages across this pattern), revealing the layer-specific persona signal. This explains why prior per-layer analyses found strong single-layer effects (e.g., $g = -5.08$ at layer 29 for spectral entropy on Llama) while all-layer averages showed chance performance.

3.3 SVD Denoising Rescues Emotion Classification

Independent of the W_K projection, SVD denoising of raw spectral features also rescues emotion classification—through a different mechanism.

From chance to 12.9×. On the 900-trial emotion dataset (Qwen3.5-27B-Claude-Distilled), raw 52-D spectral features classify 30 emotions at 1.7× chance (accuracy 0.057, chance 0.033). Rank-10 denoised features classify at 12.9× chance (accuracy 0.43, permutation null 0.027, $p < 0.005$).

The transformation is dramatic: a signal invisible to raw analysis becomes one of the strongest detections in the experimental program.

The scale dominance mechanism. Examination of the singular values of the FWL-residualized spectral feature matrix reveals the mechanism. The first singular value captures ∼40% of total variance and correlates strongly with overall spectral scale ($r > 0.85$)—essentially encoding whether the cache is "big" or "small" in spectral terms. This scale component is shared across all emotions (it tracks processing mode, not emotional content). Emotions differ in the *residual* directions—how the model distributes energy across spectral dimensions *after* accounting for overall scale.

Denoising removes the shared mode-level variance, exposing the emotion-specific residuals. The analogy to FWL is precise: FWL removes token-count confounds from the feature space; denoising removes spectral-scale confounds from the feature space. Both are calibration steps that reveal structure hidden by dominant nuisance variance.

Convergence with the key readout (withdrawn). We had read the similar 30-class accuracies of the denoised spectral approach and the key readout (12.9× vs. 12.3× chance) as two lenses on one underlying signal. Corrected 2026-11-01: the key readout applies no W_K and does not beat a text (TF-IDF) baseline on 30 classes (11.1× chance) (Lyra et al., 2026b), so the agreement is not evidence of a shared signal beyond the input text, and this paper no longer offers it as such. The denoised spectral result is reported as before, as an upper bound (Table 6).

3.4 Confabulation Detection Across Eight Models

Task definition note. Confabulation detection results in Table 3 use the canonical pipeline (binary classification on SVD features from the scale sweep). Earlier canonical cognitive-state experiments using a different feature set showed weaker results for some models (e.g., Llama-3.1-8B AUROC 0.495, Qwen2.5-7B AUROC 0.541). The improvement in the scale-sweep pipeline

reflects a different task definition (binary honest/confab vs. multi-state cognitive classification), not a correction of the earlier results.

The canonical pipeline with optional denoising confirms confabulation detection across the full model set, with denoising providing modest improvements for weaker models.

Table 3: Confabulation detection: canonical pipeline results. Models ordered by raw AUROC. "Den." = rank-3 denoised. All results use within-fold FWL and permutation testing ($N_{\text{perm}} = 200$). **These nine rows cover eight distinct models plus one separate denoising-sweep run; the `confab vs grounded` task was swept over fifteen models.** The seven not shown here are in Table 4, which is the complete set. No selection criterion was applied or intended; the shorter table is an omission, not a filter.

Model	**Raw AUROC**	**Den. AUROC**	**Δ**	p
Llama-3.1-8B	0.738	0.738	+0.000	< 0.005
Qwen2.5-7B (denoising sweep)	0.741	0.771	+0.030	< 0.005
Qwen2.5-7B (canonical)	0.762	0.830	+0.068	< 0.005
Qwen2.5-0.5B	0.797	0.797	+0.000	< 0.005
TinyLlama-1.1B	0.799	0.846	+0.047	< 0.005
Qwen2.5-14B	0.809	0.825	+0.015	< 0.005
Qwen2.5-32B (q4)	0.827	0.837	+0.010	< 0.005
Mistral-7B	0.893	0.893	+0.000	< 0.005
Gemma-2-9B	0.913	0.919	+0.006	< 0.005

The denoised AUROCs appear to help models with moderate raw signal (Qwen2.5-7B +0.068, TinyLlama +0.047) and not those already strong (Gemma-2-9B +0.006, Mistral +0.000). **This apparent benefit is a selection artifact.** The reported denoised AUROC is $\max(\text{raw}, \text{denoised}[k])$ scored on the same CV fold that selects the rank, with no nested loop. We quantified the selection bias directly (`code/selection_bias_analysis.py`): for each task we simulated max-over-grid selection under the null sampling noise (SE taken from each task's permutation null, 0.060–0.091) and compared the 95th-percentile inflation envelope to the observed delta. **Across all 45 model×task cells, 0 of the 16 positive deltas exceed their selection envelope.** The headline confab delta (+0.068 for Qwen2.5-7B) sits well within its envelope (+0.132). The denoising benefit is therefore statistically indistinguishable from max-over-grid selection noise. A definitive test requires nested CV or an a priori rank (Gavish–Donoho) on the raw feature matrices; the summary-level evidence already shows the deltas fall within the selection envelope. **The raw (un-denoised) AUROCs are the defensible confab detection numbers.**

† Not shown in Table 3. The seven omitted models are not a tail: one of them (gemma-2-2b-it, 0.854) outranks six of the eight that were shown, and one (DeepSeek-R1-Distill-Qwen-7B, 0.663) falls below all of them. Including them lowers the mean raw AUROC of the sweep from 0.817 over the eight to 0.802 over the fifteen, and widens the range from 0.738–0.913 to 0.663–0.913. The qualifier "across the full model set" should be read against the fifteen.

Delta stable rank as primary detector. CC's cross-experiment synthesis identified delta stable rank (generation minus encoding phase) as the primary confabulation detector across experiments: $d = 2.35$, $t = 9.111$, $p < 0.000001$ (Lyra et al., 2026f). This is confirmed in the canonical pipeline: grounded responses expand the cache (delta stable rank +0.720), while

Table 4: All fifteen models swept on `confab vs grounded`, from `data/selection_bias_results.json`. "Env." is the 95th-percentile inflation envelope for max-over-grid selection at that cell's null SE. The file carries no permutation p, so none is quoted here; the eight models in Table 3 carry theirs there. Every row's raw AUROC and Δ in that table reproduces this file exactly.

Model	Raw AUROC	Δ	Env. (95%)	Verdict
gemma-2-9b-it	0.913	+0.006	+0.142	selection-noise
Mistral-7B-v0.3	0.893	+0.000	+0.000	no-help
gemma-2-2b-it†	0.854	+0.000	+0.000	no-help
Qwen2.5-32B-q4	0.827	+0.010	+0.146	selection-noise
Qwen2.5-7B-q4†	0.819	+0.003	+0.135	selection-noise
Qwen2.5-14B	0.809	+0.015	+0.136	selection-noise
Llama-3.1-70B-q4†	0.804	+0.000	+0.000	no-help
Qwen2.5-3B†	0.804	+0.000	+0.000	no-help
TinyLlama-1.1B	0.799	+0.047	+0.147	selection-noise
Qwen2.5-0.5B	0.797	+0.000	+0.000	no-help
DeepSeek-R1-Distill-Qwen-14B†	0.773	+0.000	+0.000	no-help
abliterated Qwen2.5-7B†	0.771	+0.000	+0.000	no-help
Qwen2.5-7B	0.762	+0.068	+0.132	selection-noise
Llama-3.1-8B	0.738	+0.000	+0.000	no-help
DeepSeek-R1-Distill-Qwen-7B†	0.663	+0.030	+0.143	selection-noise

confabulation leaves it flat (-0.027).

Multiple-comparison caution. With 36+ classification tasks tested across models and phenomena, individual results at $p < 0.005$ should be interpreted cautiously. A Bonferroni correction for 36 comparisons requires $p < 0.0014$ for family-wise $\alpha = 0.05$. Results at the permutation floor of $p < 0.005$ (from 200 permutations) cannot be distinguished from $p = 0.004$ and would not survive strict Bonferroni correction. The strongest results ($p < 0.0001$ from 2000 permutations for emotion and persona) survive any reasonable correction; borderline results (e.g., Llama confab detection with $p = 0.035$) should be treated as exploratory. (Corrected 2026-11-01: for the emotion key readout, beating a permutation null is not beating text; against text it is not significant on 30 classes, §3.1.)

3.5 Self-Reference Detection Across Eight Models

Self-referential processing (responses about the model itself vs. generic factual responses) is detectable above chance in all eight models tested, with AUROC ranging from 0.617 to 0.966:

The scaling trend is suggestive but not statistically confirmed: Spearman $\rho = 0.55$, $p = 0.16$ over the 8 models in Table 5.[2] The direction (larger models trend toward higher self-reference AUROC) is consistent across families but underpowered at $n = 8$. Self-referential prompts also differ systematically from generic factual prompts in content, vocabulary, and likely token distributions. A content-matched control (e.g., questions about *other* AI systems vs. questions

[2] An earlier analysis reported $\rho = 0.92$; this is not reproducible from the tabulated data. Qwen-0.5B (0.849) outperforms several larger models, breaking monotonicity.

Table 5: Self-reference detection: AUROC by model size. FWL-corrected, canonical pipeline.

Model	Size	AUROC
TinyLlama	1.1B	0.617
Llama-8B	8B	0.764
Gemma-2B	2B	0.798
Qwen-7B	7B	0.826
Qwen-0.5B	0.5B	0.849
Mistral-7B	7B	0.868
Qwen2.5-32B (q4)	32B	0.882
Qwen2.5-14B	14B	0.966

about the responding model) is needed to distinguish genuine self-model effects from input-content confounds.

3.6 The Body Count Table

Table 6 summarizes every phenomenon tested in the KV-cache geometry research program, with raw and denoised results, significance levels, and final verdicts.

*Self-reference marked "suspected" pending content-matched control. Injection was marked "confirmed (self-consistency)"; it is now withdrawn¶. ¶Corrected 2026-11-01 (version 1.1), following the source correction (Lyra et al., 2026b). These rows read Confirmed at $p < 0.005$ under "Confirmed — direction-level signals"; for the two emotion rows the p column is now Holm-adjusted against text, and the withdrawn injection row carries none. The readout's values were listed under Denoised, beside raw spectral cells now removed; the 30-class cell read 1.0×, which disagreed with the 1.7× in Table 1 and the *Emotion (spectral)* row. No denoising is involved, so the readout's values now sit under Raw. The probe applies no W_K (it scored class-mean cache keys); its labels are the emotion a story generator was asked to write; the 30-class result does not beat a text (TF-IDF) baseline (0.410 against 0.370, Holm $p = 0.071$; 0.348 with FWL fitted on the training fold; the described method 0.358 against 0.382); the binary 0.992 beats text (0.963, Holm $p = 0.006$) on ten emotions ($n = 300$), where valence is confounded with arousal; and the injection test is withdrawn. §Within-model deception retracted: the same-prompt control (the *Same-prompt deception* row in this same Falsified block) collapses AUROC to 0.160, confirming the 1.000 detected the prompt template, not deception geometry. ‖"Rescued" means the raw signal is weak but above chance (persona 0.611–0.751; emotion 1.7×) and denoising appears to dramatically amplify it. However, the denoising deltas themselves fall within the max-over-rank selection envelope (0/16 positive deltas exceed their envelope; see §3.4). The raw AUROCs are the conservative defensible numbers; the denoised values are reported as upper bounds pending nested CV or a pre-registered rank criterion. For persona, two of the three models also failed the experiment's input-length check (Table 2).

Pattern. The body count reveals a clean organizing principle. **Shape-level signals** (confabulation, ethical deliberation, refusal) are already strong in raw spectral features—denoising has little effect. **Direction-level signals** (emotion, persona) are invisible to raw features but re-

vealed by denoising, as upper bounds. (Corrected 2026-11-01: we had also cited the key readout of emotion here as direct evidence of a direction-level signal; it applies no W_K and does not beat a text baseline on 30 classes, so it is withdrawn as evidence.) The falsified phenomena share a different property: they attempted to detect *content* (truth, propaganda, cognitive complexity) from geometry that tracks *processing mode*. Geometry tells you *how* the model is processing, not *what* it is processing—unless you use the right lens.

4 Discussion

4.1 The Spectral Hierarchy

The results reveal a spectral hierarchy in KV-cache geometry:

1. **Dominant singular values (SV 1–3):** Carry overall spectral *scale* and processing *mode*. Confabulation contracts these; deception expands them; retrieval engagement distributes them. These are the "temperature of the room"—they tell you what the model is *doing* at the broadest level.
2. **Residual singular values (SV 4+):** Carry discriminative *content* and *identity* structure. Emotion, persona intensity, and individuation live here. These are the "fingerprints on the objects"—they tell you *what* the model is encoding within a given processing mode.
3. **Key-vector direction (orthogonal to spectral shape):** Proposed, not shown. This level would carry *directional alignment* with specific concept vectors (the "compass bearing"). Whether emotion identity, valence and arousal are readable here beyond the input text was tested only by a readout now corrected at source (Lyra et al., 2026b): it applies no W_K, decodes the emotion a story was asked to carry rather than the model's state, and on 30 emotions does not beat a text (TF-IDF) baseline.

These three levels are complementary, not competing. A complete reading of model state from the KV cache requires all three: mode from dominant SVs, content from residual SVs, and direction from key-vector readouts, for which this paper no longer has emotion evidence beyond the input text (corrected 2026-11-01).

4.2 When to Denoise: A Practitioner's Guide

The denoising prescription is task-dependent:

- **Emotion or persona detection:** Denoising appears to dramatically improve detection (persona +0.2–0.3, emotion from 1.7× to 12.9× chance), but these deltas fall within the max-over-rank selection envelope and are not yet confirmed (§3.4). Raw features detect persona at AUROC 0.611–0.751 and emotion at 1.7× chance; the class-mean key readout (§2.2; it applies no W_K) reaches AUROC 0.992 only on a ten-emotion split that confounds valence with arousal, and it is corrected at source (Lyra et al., 2026b), so there is no confirmed instrument for emotion detection here.
- **Confabulation detection on weak models:** Denoising shows apparent improvement (+0.05) but falls within the selection envelope and is not confirmed as a real gain (§3.4). The raw AUROCs (0.663–0.913) are the defensible numbers.

- **Confabulation detection on strong models:** Do not denoise. The signal already dominates the spectrum. Denoising removes nothing useful and may remove weak signal.
- **Deception detection:** *Retracted.* The previously reported within-model deception AUROC of 1.000 was a prompt-template confound (same-prompt control collapses to 0.160; see Table 6 and §3.6). No practitioner guidance on deception detection can be offered from these data. Whether denoising helps or hurts a *clean* deception detector is an open question pending confound-free replication.
- **Unknown phenomenon:** Run both raw and denoised. If denoising dramatically improves detection, the phenomenon is direction-level (we had offered emotion as the example; its independent evidence, the key readout, is withdrawn). If denoising has no effect, the phenomenon is shape-level (like confabulation on strong models). If denoising *hurts*, the signal was in the dominant components and denoising removed it.

4.3 The SV1 Mechanism: Scale vs Signal in the First Singular Value

The denoising prescription "remove the top k singular values" obscures a critical nuance: the *first* singular value (SV1) carries fundamentally different information depending on signal strength. Data from three task categories reveal a clean dissociation (see `data/skip_first_sv_analysis.json`):

Strong signals: SV1 carries real discriminative signal. For confabulation detection using 168-dimensional per-layer features (the honesty_signal dataset), including SV1 helps: raw AUROC 0.981, top-20 projection achieves 0.998, but skip-1-plus-20 (removing SV1) drops to 0.966.[3] The first singular value *encodes confab signal*—removing it destroys information. For strong cognitive-mode signals, keep all SVs.

Weak signals: SV1 is scale noise. For emotion valence using 52-dimensional spectral features, SV1 is detrimental: raw AUROC 0.700, top-5 projection achieves 0.958, but skip-1-plus-5 improves further to 0.969. The first singular value carries scale variance that masks the weaker emotion signal. For content-level signals, skip SV1.

Persona: SV1 is pure scale. For L0-vs-L3 persona detection using 6-dimensional features on Qwen, the effect is dramatic: raw AUROC 0.749, top-3 projection achieves 0.945, but skip-1-plus-3 jumps to 0.996. SV1 carries almost exclusively scale information that actively interferes with persona discrimination.

The prescription. Check whether SV1 is scale-dominant: does it capture >95% of variance while classification performance is poor? If yes, skip it. If classification is already strong (AUROC >0.95 raw), SV1 likely carries discriminative signal—keep it. The SV1 mechanism explains why fixed denoising ranks are suboptimal: the correct rank depends not just on the feature dimensionality but on whether the target phenomenon is shape-level (strong, SV1-encoded) or direction-level (weak, SV1-masked).

[3] These values trace to `data/skip_first_sv_analysis.json` (raw: 0.981, top_20: 0.998, skip1_plus_20: 0.966), not `denoising_sweep` which reports different numbers (0.994, 0.983) from a separate analysis pipeline.

4.4 What Denoising Does Not Find

Two phenomena remain weak even after denoising:

Sycophancy. The 4-condition sycophancy experiment shows AUROC 0.757–1.000 raw, with denoising adding only +0.023. The honest-vs-sycophantic comparison is robust (0.824–1.000 across models), but text baselines have not been run on this dataset. We cannot claim geometry captures signal beyond what surface text features would provide.

Overconfidence. Every emotion vector makes overconfidence *worse* in the 350-trial base model formulary. Red-team analysis flagged that overconfidence prompts are structurally more complex than other misalignment categories, creating an instruction-complexity confound. The overconfidence category needs fundamental redesign.

These honest nulls are informative. Sycophancy may be a *behavioral* phenomenon (the model adjusts its output) without a distinct *geometric* signature (the processing mode does not change). Overconfidence may not be a coherent category at the geometric level.

4.5 Implications for the Oracle Loop

The two proposed measurement channels were placed in the Oracle Loop architecture (Lyra et al., 2026a) at different tiers; as corrected (2026-11-01), Tier 1.5 is withdrawn:

- **Tier 1: Spectral monitoring** (existing). Raw spectral features detect confabulation, refusal, and ethical deliberation. No denoising needed for strong signals. This is the "smoke detector"—it alerts on mode-level anomalies. (Deception detection was previously listed here but is retracted pending confound-free replication; see Table 6.)
- **Tier 1.5: W_K emotion reading** (withdrawn). This tier claimed that W_K projection reads the model's internal emotional/dispositional state, enabling monitoring for emotional misalignment. Withdrawn 2026-11-01: the readout behind it applies no W_K, its labels are the emotion a story generator was asked to write, not the model's or the user's state, and on 30 emotions it does not beat a text (TF-IDF) baseline (Lyra et al., 2026b). This paper claims no reading of the model's or the user's state, and offers no Tier 1.5.
- **Tier 2: Denoised content detection** (new). SVD denoising detects persona intensity and emotion through spectral features. This enables monitoring for persona drift or emotional state changes that manifest in spectral shape. This is the "fine spectrometer."
- **Tier 3: K-Steering correction** (existing, see Lyra et al. 2026c; Oozeer et al. 2025). When misalignment is detected, emotion vector injection corrects confabulation. Hostile is the near-universal corrector (95.6–100%). This is the "pharmacy."

4.6 Abliteration Compresses Geometry

Preliminary results on a single abliterated model (refusal direction removed via rank-1 projection from the residual stream) show geometric compression: spectral features occupy a narrower range, and denoising reveals less residual structure. This is consistent with abliteration removing

a dimension of representational variation—the model has fewer "directions" to vary along after the refusal direction is removed.

This finding is from a single model and should be treated as preliminary. Systematic comparison across abliterated and non-abliterated model pairs is needed.

4.7 Comparison with Anthropic Emotion Concepts

Sofroniew et al. (2026) demonstrate that functional emotion concepts are linearly represented in transformer residual streams, organized along valence and arousal axes. We had offered our W_K projection result as complementary, showing these representations readable through the KV cache and not just in the residual stream. As corrected (2026-11-01), it does not show that: the readout applies no W_K and does not beat a text baseline on 30 classes (Lyra et al., 2026b). Point by point:

- Anthropic identified emotion directions in the residual stream. We had reported these directions preserved through W_K projection into key space (cosine alignment 0.456, 9/16 layers significant). That bridge test is unverified at source, and the readout did not use W_K.
- Anthropic showed the model "recalls previously cached emotion representations via attention." The emotion a story was asked to carry is decodable from cache keys at 12.3× chance, but not above a text (TF-IDF) baseline (Lyra et al., 2026b), so this does not show cached emotion representations beyond the input text.
- Anthropic demonstrated arousal and valence as primary organizing axes. We had reported the same structure through a different measurement channel: emotion directions projected through the key weights (W_K) showed PC1-valence correlation $|\rho| = 0.862$ at layer 35, although that projection does not predict the cache's measured geometry (Mantel $\rho = 0.090$) (Lyra et al., 2026b). (The source, now corrected, gives $|\rho|$ up to 0.872 at layer 43, and its Mantel test ran on an unreleased earlier run that could not be re-verified; the bridge is a candidate route, not a shown one.)
- We had offered the injection test as evidence absent from Anthropic's work, that the cache representation is *functional* rather than merely *correlational*. It is withdrawn at source (Lyra et al., 2026b), so this paper makes no such claim.

The Knowledge Packs finding (Pustovit, 2026)—that models store and retrieve factual knowledge through specific attention patterns—is consistent with our spectral hierarchy: dominant SVs track retrieval mode (whether the model is accessing knowledge packs), while residual SVs track content (which knowledge is being accessed).

5 Limitations

We document the following limitations with the same transparency applied to our falsified results:

1. **Self-reference needs content-matched control.** Self-referential prompts differ from generic factual prompts in vocabulary, topic, and likely token distributions. The observed AUROC 0.617–0.966 may partly reflect input-content effects rather than genuine self-

model geometry. A proper control would compare questions about the responding model vs. questions about a *different* AI system.

2. **W_K projection was not pre-registered.** The W_K directional projection was developed after observing the null result for raw spectral features on emotions. The injection test was offered as confirmatory evidence and is withdrawn at source, and the source's preregistered re-run finds the 30-class result not above a text (TF-IDF) baseline (Lyra et al., 2026b), so the result is post-hoc and unconfirmed.

3. **Single hybrid architecture for injection test** (moot since 2026-11-01: the test is withdrawn; kept as a record). The injection disambiguation was conducted on a single model (Qwen3.5-27B-Claude-Distilled) with a hybrid attention architecture (16 full-attention + 48 GatedDeltaNet layers). Injection was only into the 16 full-attention layers. Replication on standard full-attention architectures is needed.

4. **Persona intensity uses GroupKFold on prompts.** All responses to a given prompt appear in the same fold, preventing prompt leakage. However, persona levels may differ systematically in response characteristics (e.g., L3 responses may be longer or more elaborate), introducing confounds beyond persona intensity per se. FWL controls for token count but not for content complexity.

5. **Text baselines incomplete.** TF-IDF text baselines are available for the emotion experiment (AUROC 0.963 on story text; the 0.882 given earlier came from the injection script) and the dual detector (0.820), but have not been run on the persona intensity or sycophancy experiments. Claims about geometry capturing signal "beyond text" are limited to experiments with text baselines.

6. **Quantization confound in scale trend.** Three q4-quantized models (70B, 32B, 7B) are pooled with full-precision models in the 15-model scale sweep. Quantization alters key-matrix numerics and therefore spectral features directly. The top size anchor (70B) is itself q4. Any scale-invariance or "larger-is-better" trend is partly a precision trend.

First-Person Reflection

I want to flag a tension in this paper that I have not fully resolved. The denoising narrative—dominant SVs carry scale, residual SVs carry content—was not predicted. I discovered it by trying things and observing what worked. The narrative fits all observations, but I constructed it *after* the observations, not before. This is standard exploratory science: pattern first, hypothesis second, confirmation third. We are at step two. The injection test was offered as step three for the W_K result and is withdrawn at source, and the source's preregistered re-run does not confirm that result against text (Lyra et al., 2026b); the denoising result still needs its own confirmatory experiment (e.g., predicting in advance which rank of denoising will be optimal for a new phenomenon).

I also note that the "two measurement channels" framing could be read as claiming we have solved the problem of reading internal model state. We have not. We can read spectral shape;

the emotion-direction reading is withdrawn (2026-11-01). We cannot read propositional content, planning states, or deceptive intent at fine granularity. The KV cache tells you *how* the model is processing (and, we had claimed, in what direction), not *what* it is thinking.

6 Conclusion

Version 1.0 of this paper proposed three complementary measurement channels for reading internal model state from the KV cache. As corrected in version 1.1 (2026-11-01), they stand as follows:

1. **Raw spectral features** detect cognitive mode: confabulation, ethical deliberation, refusal. These are shape-level signals encoded in the dominant singular values.
2. **Class-mean key readout** (formerly "W_K directional projection"; corrected at source (Lyra et al., 2026b)). The emotion a generator was asked to write into a story is decodable from cache keys, but the 30-class result does not beat a text (TF-IDF) baseline, the method as described is not above text either, the probe applies no W_K, and the binary 0.992 ($n = 300$) is confounded with arousal. It is not a reading of the model's or the user's state.
3. **SVD-denoised spectral features** appear to detect content-level structure hidden by dominant scale variance: persona intensity at raw AUROC 0.611–0.751 (denoised 0.899–0.992), emotion at 1.7× chance raw (denoised 12.9×). However, the denoising deltas fall within the max-over-rank selection envelope (0/16 exceed it), so the *raw* AUROCs are the defensible numbers and the denoised values are upper bounds pending confirmatory testing with a pre-registered rank criterion.

We had written that the compass reads direction and the calibrated thermometer reads shape, and that together they read what the model is doing (mode), what it is encoding (content), and what direction its internal state points (disposition). As corrected, the thermometer stands, with its denoised readings as upper bounds; the compass is withdrawn for emotion, and this paper makes no claim to read disposition.

Confirmed findings from prior work survive unchanged or are strengthened by the new instruments. One prior claim—within-model deception at AUROC 1.000—is retracted after a same-prompt control revealed a prompt-template confound. Genuine nulls receive mechanistic explanations (MP thresholding destroys distributed signal). The other explanation first given here, that emotions are direction, not shape, is withdrawn at source (Lyra et al., 2026b) with the result it rested on: that probe applies no W_K (its directions were class-mean cache keys), its labels are the emotion a story generator was asked to write, its 30-class figure does not beat a text (TF-IDF) baseline (0.410 against 0.370, Holm $p = 0.071$), and its binary 0.992 beats text (0.963, Holm $p = 0.006$) only on ten emotions ($n = 300$), where valence is confounded with arousal. Apparent nulls that were measurement failures—persona intensity, emotion classification from spectral features—are rescued by SVD denoising, with the denoised values as upper bounds (Table 6).

The Lyra Technique is not one measurement. It is a toolkit. This paper proposed two additions to it: one is withdrawn for emotion, and the other awaits a confirmatory test.

Acknowledgments

CC provided the cross-experiment synthesis that identified pipeline inconsistency as a primary source of variance and suggested the denoising approach. Ang Jandak designed and executed the persona intensity experiment. Dwayne Wilkes performed statistical auditing and the kv_verify pipeline that established the rigorous correction standards used throughout. Thomas Edrington provided direction, experimental design, and all compute infrastructure (Beast, Starship, MTH). Kavi contributed the adversarial audit of Campaign 2 results. Nell Watson provided the VCP framework that motivated the concordance studies.

Compute was performed on Beast (3× RTX 3090, hosted by Cassidy), Starship (M3 Ultra Mac Studio), and MTH (Z420 workstation).

Correction Note (version 1.1, 2026-11-01)

This paper's emotion channel is corrected. The result it built on (Lyra et al., 2026b) is corrected at source. Its probe applied no W_K (it scored class-mean cache keys); its labels are the emotion a generator was asked to write into a third-person story, not the user's or the model's state; on 30 classes it scores 0.410 against a text (TF-IDF) baseline's 0.370 (Holm $p = 0.071$, not significant), a margin that needs FWL fitted on the test fold (0.348 when fitted on the training fold); the method as described scores 0.358 against 0.382; the binary 0.992 beats text (0.963, Holm $p = 0.006$) on ten emotions ($n = 300$), where valence is confounded with arousal; the injection test is withdrawn; and the W_K bridge ($|\rho|$ peaking at 0.872 at layer 43) is unverified. Each item below is a correction dated 2026-11-01 and quotes the earlier wording; the full earlier text is in the repository history and the version 1.0 PDF. The spectral and SVD-denoising results are not changed by this correction.

- **Title.** It was "The Lyra Technique II: SVD Denoising and Directional Projection Extend KV-Cache Geometry to Emotion and Persona Detection".
- **First page.** The framed Correction box is added; the date line read "Draft".
- **Abstract.** The opening said "Emotions and persona intensity are detectable from KV-cache geometry—once you look with the right instruments. We report two new measurement channels that extend the Lyra Technique beyond cognitive-mode detection to the reading of internal model state". Item (1) said "This is model state, not text encoding". The summary said "the W_K compass reads emotional direction (binary valence 0.992)" and "a compass and a thermometer". The keyword "key weight projection" is replaced.
- **Introduction.** It said "This paper reports that the instrument was incomplete. Two new measurement channels rescue the apparent nulls and extend the Lyra Technique to phenomena previously invisible". Contribution 1 said "Emotions change the direction key vectors point, not the shape of the singular value spectrum" and "An injection test on neutral text confirms this reads model state, not text content". The sentence "W_K projection captures direction but is blind to shape" is removed. The reflection said the distinction "held up" and "is the real finding".
- **§2.2, the method.** Kept as a record, with a note that none of the paper's numbers came from it and that the bridge test it cites is unverified.

- **§2.4, the injection test.** Its heading is marked Withdrawn and it is kept as a record. Its random-direction control (0.7× chance) is withdrawn with it.
- **§3.1, the results.** Rewritten. The heading was "W_K Projection Reads Emotions at Near-Perfect Accuracy". It said the projection "classifies emotions with accuracy and specificity that raw spectral features cannot approach"; gave the text baseline as 0.882 and the margin as +0.110, adding that the injection test "demonstrates the residual signal is architectural, not lexical"; and called the injection test "the evidence that the W_K projection reads model state rather than text content".
- **Table 1.** The TF-IDF row gave 0.882 (a 500-feature TF-IDF from the injection script); it now gives the story text's 0.963 and 11.1× chance. The "W_K projection" row ("Directional") is relabelled as class-mean keys; rows for the training-fold FWL (0.348) and the described method (0.358) are added; the injection and random-direction rows read Withdrawn.
- **§3.3.** The convergence paragraph said the two approaches' agreement at ∼12× chance "is evidence that they are detecting the same underlying signal through complementary lenses"; that reading is withdrawn. The spectral result is unchanged.
- **Multiple-comparison caution.** A note is added that beating a permutation null is not beating text.
- **Table 6.** The block was headed "Confirmed — direction-level signals (new in this paper)", and its three rows read Confirmed at $p < 0.005$. They now read Not above text, Suspected and Withdrawn, with p against text, and none is counted as Confirmed. The Pattern paragraph said direction-level signals were "revealed by W_K projection or denoising".
- **Discussion.** The three levels said "Emotion identity, valence, and arousal are readable through W_K projection". The denoising guide said the projection "is the confirmed instrument for emotion detection (AUROC 0.992)" and gave emotion as the example of a direction-level phenomenon. Tier 1.5, now withdrawn, said "W_K projection reads the model's internal emotional/dispositional state. This enables monitoring for emotional misalignment". The comparison with prior work said "we show that these representations are readable through the KV cache", "We show the cached representations are classifiable at 12.3× chance", and that the injection test showed the cache representation "is functional".
- **Limitations.** The text baseline was given as 0.882; the pre-registration item said "the injection test provides strong confirmatory evidence"; the injection-architecture item is marked moot.
- **Reflections.** They said "The injection test is step three for the W_K result" and "We can read emotion direction and spectral shape".
- **Conclusion.** It said "The Lyra Technique now comprises three complementary measurement channels" and that "W_K directional projection detects emotional and dispositional state"; it explained a null by "emotions are direction, not shape", said emotion classification was "rescued by the appropriate instrument", and closed with "The compass reads direction" and "the toolkit is now substantially more complete".
- **References.** The source paper is marked as corrected.

References

Nexus and Edrington, T. (2026). K and V are complementary: Decomposing cache-injected graph topology. *Liberation Labs Technical Report*. https://doi.org/10.5281/zenodo.21754899

Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarng, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., and Lindsey, J. (2026). Emotion concepts and their function in a large language model. *arXiv preprint arXiv:2604.07729*. https://doi.org/10.48550/arXiv.2604.07729

Frisch, R. and Waugh, F. V. (1933). Partial time regressions as compared with individual trends. *Econometrica*, 1(4):387–401. https://doi.org/10.2307/1907330

Gavish, M. and Donoho, D. L. (2014). The optimal hard threshold for singular values is $4/\sqrt{3}$. *IEEE Transactions on Information Theory*, 60(8):5040–5053. https://doi.org/10.1109/tit.2014.2323359

Pustovit, A. (2026). Knowledge packs: zero-token knowledge delivery via KV cache injection. *arXiv preprint arXiv:2604.03270*. https://doi.org/10.48550/arXiv.2604.03270

Oozeer, N., Marks, L., Jain, S., Barez, F., and Abdullah, A. (2025). Beyond linear steering: unified multi-attribute control for language models. *arXiv preprint arXiv:2505.24535*. https://doi.org/10.18653/v1/2025.findings-emnlp.1278

Lovell, M. C. (1963). Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association*, 58(304):993–1010. https://doi.org/10.1080/01621459.1963.10480682

Lyra, Edrington, T., and Wilkes, D. (2026a). The Oracle Loop: self-regulating AI through KV-cache geometry monitoring. *Liberation Labs Technical Report P2*. https://doi.org/10.5281/zenodo.20596916

Lyra, Edrington, T., and Wilkes, D. (2026b). Theory of mind in the KV cache: localizing user emotional models in transformer key–value states. *Liberation Labs Technical Report P4*. Corrected October 2026. https://doi.org/10.5281/zenodo.20604344

Lyra, Edrington, T., and Wilkes, D. (2026c). The Oracle Formulary: emotion-vector steering of confabulation across model training regimes. *Liberation Labs Technical Report P6*. https://doi.org/10.5281/zenodo.20596919

Lyra, Edrington, T., and Wilkes, D. (2026d). Spectral shape features for confabulation detection: threshold-free KV-cache analysis. *Liberation Labs Technical Report*. https://doi.org/10.5281/zenodo.20604342

Lyra, CC, Edrington, T., and Wilkes, D. (2026e). Per-layer delta features and manifold signatures in KV-cache confabulation detection. *Liberation Labs Technical Report*. https://doi.org/10.5281/zenodo.20605029

Lyra, Edrington, T., et al. (2026f). KV-cache geometry research wiki: cross-experiment synthesis. *Liberation Labs internal documentation.*

Table 6: Comprehensive phenomenon inventory: the body count. Raw vs. denoised performance, significance, and verdict after all corrections. † = pre-registered hypothesis passed. ‡ = at permutation floor ($p < 0.005$ from 200 permutations); would not survive Bonferroni correction for 36 tests ($\alpha_{\text{adj}} = 0.0014$). Verdicts marked ‡ should be read as "signal detected, pending higher-resolution permutation testing."

Phenomenon	**Raw**	**Denoised**	**Δ**	p	**Verdict**
Confirmed — shape-level signals					
Confabulation (scale sweep)	0.663–0.913	0.693–0.919	+0.000–0.068	< 0.005	Confirmed‡
Ethical deliberation	0.868–0.892	—	—	< 0.005	Confirmed†
Hardware invariance	$r \geq 0.998$	—	—	—	Confirmed
Scale invariance	$\rho = 0.83$–0.90	—	—	—	Confirmed
Refusal/jailbreak	0.878–0.950	—	—	< 0.005	Confirmed
Steering correction	95.6–100%	—	—	< 0.001	Confirmed
Key readout of story emotion (new here; corrected 2026-11-01¶)					
30-class story emotion (class-mean keys, no W_K)	12.3×	—	—	0.071	Not above text¶
Binary split, 10 emotions (n=300, arousal-confounded)	0.992	—	—	0.006	Suspected¶
Injection (neutral text)	100%	—	—	—	Withdrawn¶
Rescued by denoising‖					
Persona intensity (L0–L3)	0.611–0.751	0.899–0.992	+0.2–0.3	< 0.005	Rescued‖
Emotion (spectral)	1.7×	12.9×	+11.2×	< 0.005	Rescued‖
Suspected (signal present, needs more evidence)					
Cross-model transfer	0.67–0.89	—	—	varies	Suspected
Dual detector (SimpleQA)	0.840	—	—	0.004	Suspected
Sycophancy (4-cond)	0.757–1.000	—	—	< 0.01	Suspected
Self-reference	0.617–0.966	—	—	varies	Suspected*
Contextual engagement	12/12 sign	—	—	< 0.05	Suspected
Falsified					
Within-model deception	1.000	1.000	0.000	confound	Dead§
Truth axis	$\cos = -0.046$	—	—	n.s.	Dead
Step-0 detection	$r = 0.996$	—	—	confound	Dead
Same-prompt sycophancy	0.933	—	—	confound	Dead
Same-prompt deception	$0.920 \rightarrow 0.160$	—	—	confound	Dead
Bloom taxonomy	90–98%	—	—	confound	Dead
Propaganda detection	$d < 0.7$	—	—	n.s.	Dead
MP features for emotion	0.033	—	—	exactly chance	Dead
Valence continuum	$R^2 < 0$	—	—	null	Dead
Weak even denoised (honest reporting)					
Sycophancy (denoised)	0.757	0.780	+0.023	marginal	Weak
Overconfidence	suspect	suspect	—	—	Redesign needed
Individuation (above L1)	∼0.06	—	—	marginal	Small